MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 06 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A number is selected at random from the numbers 1 to 25. The probability that the selected number is prime is: **[1]**

Options: (a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{10}{25}$ (d) $\frac{7}{25}$

2. The probability that a non-leap year contains 53 Mondays is: **[1]**

Options: (a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{5}{7}$ (d) 0

3. A single letter is selected at random from the word 'PROBABILITY'. The probability that it is a vowel is: **[1]**

Options: (a) $\frac{3}{11}$ (b) $\frac{4}{11}$ (c) $\frac{5}{11}$ (d) $\frac{2}{11}$

4. A number x is chosen at random from the set $\{-3, -2, -1, 0, 1, 2, 3\}$. The probability that $|x| < 2$ is: **[1]**

Options: (a) $\frac{2}{7}$ (b) $\frac{3}{7}$ (c) $\frac{4}{7}$ (d) $\frac{5}{7}$

5. **Assertion (A):** A lot contains 12 defective pens and 132 good pens. The probability that a pen taken out at random is good is $\frac{11}{12}$.
Reason (R): Total pens $= 12 + 132 = 144$, and favourable outcomes for a good pen $= 132 \implies \frac{132}{144} = \frac{11}{12}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish. What is the probability that the fish taken out is a male fish? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability of getting: (i) a spade or a red king (ii) neither a jack nor a queen (iii) a black face card. **[3]**

8. A box contains cards numbered $3, 5, 7, 9, \ldots, 35, 37$. A card is drawn at random from the box. Find the probability that the number on the drawn card is: (i) a prime number (ii) a number divisible by 7 (iii) a two-digit number. **[3]**

9. A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random from the lot. (i) What is the probability that this bulb is defective? (ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective? **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

An education department conducted an audit across 100 secondary schools to analyze computer-based learning facilities. The survey categorized schools based on the number of working computers available for students:

Computer Range	1 – 10	11 – 20	21 – 50	51 – 100	More than 100
Number of Schools	15	25	35	18	7

A school is selected at random from the surveyed group.

Based on the above scenario, answer the following:

(i) What is the probability that the selected school has more than 50 computers? **[1]**

(ii) What is the probability that the selected school has at most 20 computers? **[1]**

(iii) What is the probability that the selected school has between 11 and 50 computers (inclusive)? **[1]**

(iv) What is the probability that the selected school has at least 21 computers? **[1]**

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